SFT Versus Benchmark Input: Audit and JSON-Presentation Check

Status: Paired inference completed by the owner; full-suite spacing ablation prepared.

Question

The name-first checkpoint clarified on 7/10 selected held-out SFT inputs, but on 0/40 expected-clarification cases in the established benchmark. Is a presentation difference involved? Neither observation establishes the cause, and the two sample sets are not matched tasks.

Confirmed Audit Findings

The generic prompt-building code and saved ten-example probe were inspected. No benchmark fixture was used to construct or rewrite a training example.

Property	Finding
System prompt	All ten match the inference default and recorded training hash a31446de...
Header and request wrapper	Same header; same current_time / initial_state / request fields
Meta-tools	Shared definitions, identical names, schemas and leading order
Productive tool metadata	All ten match the generic benchmark definitions, descriptions and schemas
Tools in this sample	Every prompt contains two meta-tools and one productive tool
JSON serialization	SFT is compact with literal Unicode; benchmark uses default spaces and ASCII escaping
Time rendering	All ten SFT clocks retain 2026-06-01T09:00:00Z; benchmark code uses datetime.isoformat()
Request phrasing	Several successful SFT cases explicitly state that information has not been supplied

For UTC, datetime.isoformat() can spell the same instant with +00:00 instead of Z. The paired check below deliberately does **not** change this spelling. It also does not vary tool counts, request wording, state content or metadata. Those are separate potential distribution differences, not established explanations.

The three ACT mistakes in the probe involve an invalid due date, unavailable checklist content and an ambiguous task reference. This is descriptive inspection of held-out SFT rows, not permission to derive training targets from benchmark failures or evidence for a general lexical-rule diagnosis.

One Controlled Inference Check

The existing probe now supports `--benchmark-json`. It selects the **same ten** validation rows in the same order and changes only JSON spacing/escaping to the existing benchmark renderer's conventions. Decoded requests, tools, schemas, state, clock strings and expected targets are unchanged.

The saved original-SFT report is required. Preflight rejects a different selection, config, dataset, adapter hash or generation setting. The expected assistant answer is never included in either model prompt. Model loading, greedy decoding, 256 new tokens and strict parsing are unchanged. No productive tool is executed.

Run from the repository root:

```
.venv\Scripts\python scripts\probe_sft_clarification.py `
  --benchmark-json `
  --reference-report tmp\name-first-heldout-clarification-10.json `
  --output tmp\name-first-heldout-clarification-benchmark-json-10.json
```

The owner completed both probes. Saved reports have matching config, dataset, adapter-weight hashes and generation settings. Inspection confirmed identical decoded input values, tool order, system messages and expected targets. All ten inputs are ASCII, so this pair changed JSON separator spacing only.

Input presentation	ACT	CLARIFY	REFUSE	Invalid/unknown
Exact compact SFT inputs	3	7	0	0
Benchmark JSON spacing	10	0	0	0

Raw outputs confirm all seven previous clarification decisions changed to productive calls; none of the new outputs mention request_clarification. For example, row 25 switched from asking for an event time to supplying 21:00 and a 30-minute duration. These are routing observations, not semantic question scores. Reports remain under ignored tmp/ and existing files were not overwritten.

This supports substantial presentation sensitivity on these ten selected inputs. It does not establish its magnitude across the benchmark or robust policy learning.

Full-Suite Spacing Ablation

The next owner-run diagnostic evaluates the same name-first checkpoint on the unchanged 150 scenarios from scenarios/v1 and scenarios/v2. Use --compact-json with the existing run-agent-lora command and a new output file:

```
$adapter =
"models\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo_name_first\adapt
er"
.venv\Scripts\python -m baymax.eval.cli run-agent-lora `
  --scenarios scenarios\v1 `
  --scenarios scenarios\v2 `
```

```
  --adapter $adapter `
  --compact-json `
  --output results\phase2\json-spacing\name-first-compact-combined-150.json
```

Only separators in the available-tool and request-context JSON change. Unicode escaping, timestamp spelling, input values, key/tool order, system prompt, checkpoint, greedy decoding, 256-token generation cap, parser, fake execution and scoring are unchanged. In particular, the renderer does not convert +00:00 to Z. Literal Unicode versus ASCII escaping is not part of this ablation.

The default renderer remains spaced. Local result metadata records the format and inference configuration. Compact mode refuses to overwrite an existing output. No retraining, target changes or rescoring of historical artifacts is involved. The full-suite result is pending; the coding agent ran only mocked unit tests. This repeatedly inspected suite is development evidence, not an untouched final test set. Compare routing macro-F1 and confusion matrices separately from keyword wording scores and task success.

Semantic Review Is Separate

Routing, response adequacy and keyword agreement must not be conflated. A valid clarification call can ask the wrong question; a useful question can use wording not recognized by keyword checks.

The proposed next measurement is a small blinded human review of boundary responses, using request/context and the general policy rather than visible keyword scores. It would report adequacy separately and preserve official scores. This audit does not implement that review, an LLM judge, or new scoring rules. The coding agent's unblinded inspection is not independent human validation.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.